

Bank Account Management System

Title : Bank Account Management System using Classes, Objects and Inheritance.

Objective / Aim : To write a Java program to implement a Bank Account Management System using classes and objects, and to demonstrate inheritance, constructors, method overriding, deposit, withdrawal, interest calculation, overdraft facility, and money transfer.

System Requirements :

Hardware

- Standard PC/Laptop
- Core i3/i5 Processor or equivalent
- 4 GB/8 GB RAM
- Keyboard and Monitor

Software

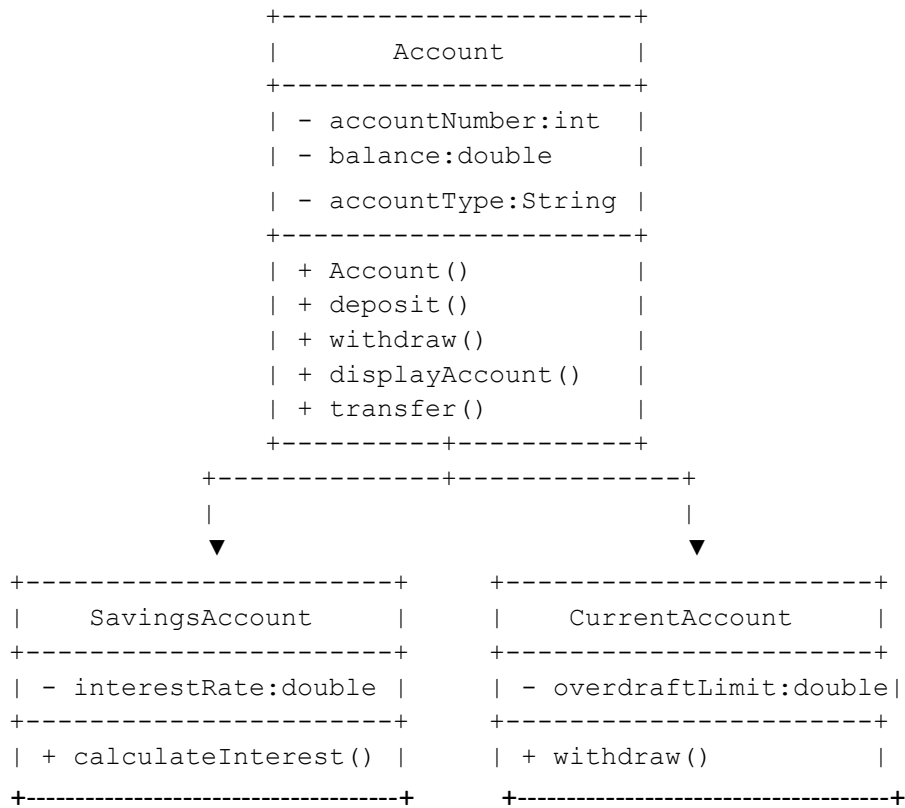
- Operating System: Windows/Linux/macOS
- Java Development Kit (JDK 17 or JDK 21)
- Eclipse / IntelliJ IDEA / NetBeans / VS Code
- Command Prompt or Terminal

Algorithm / Logic Description :

1. Start the program.
2. Create a base class named Account.
3. Declare account number, balance, and account type as data members.
4. Create a constructor to initialize the account details.
5. Define the deposit() method to add money to the account.
6. Define the withdraw() method to withdraw money if sufficient balance is available.

7. Define the transfer() method to transfer money from one account to another.
8. Create a SavingsAccount class that extends the Account class.
9. Add an interest rate to the savings account and calculate interest using the balance and interest rate.
10. Create a CurrentAccount class that extends the Account class.
11. Override the withdraw() method in CurrentAccount to provide an overdraft facility.
12. Create objects for Savings Account and Current Account.
13. Perform deposit, withdrawal, interest calculation, and transfer operations.
14. Display the updated account details.
15. Stop the program.

UML Class Diagram



Source code :

```
package core_java;

import java.util.Scanner;

class Account {

    int accountNumber;

    double balance;

    String accountType;

    Account(int accountNumber, double balance, String accountType) {

        this.accountNumber = accountNumber;

        this.balance = balance;

        this.accountType = accountType;

    }

    void deposit(double amount) {

        balance = balance + amount;

        System.out.println("Amount deposited successfully.");

    }

    void withdraw(double amount) {

        if (amount <= balance) {

            balance = balance - amount;

            System.out.println("Amount withdrawn successfully.");

        } else {

            System.out.println("Insufficient balance.");

        }

    }

    void displayAccount() {
```

```

System.out.println("Account Number : " + accountNumber);

    System.out.println("Account Type  : " + accountType);

    System.out.println("Balance      : " + balance);

}

void transfer(Account receiver, double amount) {

    if (amount <= balance) {

        balance = balance - amount;

        receiver.balance = receiver.balance + amount;

        System.out.println("Amount transferred successfully.");

    } else {

        System.out.println("Insufficient balance for transfer.");

    }

}

}

// Savings Account

class SavingsAccount extends Account {

    double interestRate;

    SavingsAccount(int accountNumber, double balance, String accountType, double interestRate) {

        super(accountNumber, balance, accountType);

        this.interestRate = interestRate;

    }

    void calculateInterest() {

        double interest = balance * interestRate / 100;

        balance = balance + interest;

    }

}

```

```
        System.out.println("Interest Rate : " + interestRate + "%");

        System.out.println("Interest Added: " + interest);
    }
}

// Current Account
class CurrentAccount extends Account {

    double overdraftLimit;

    CurrentAccount(int accountNumber, double balance,String accountType, double overdraftLimit) {

        super(accountNumber, balance, accountType);

        this.overdraftLimit = overdraftLimit;
    }

    @Override
    void withdraw(double amount) {

        if (amount <= balance + overdraftLimit)

            balance = balance - amount;

            System.out.println("Amount withdrawn successfully.");

        else

            System.out.println("Overdraft limit exceeded.");

        }

    }

// Main Class
public class BankAccountManagement {

    public static void main(String[] args) {
```

```
Scanner sc = new Scanner(System.in);

// Creating Savings Account

    SavingsAccount savings = new SavingsAccount(101, 10000, "Savings", 5);

// Creating Current Account

    CurrentAccount current = new CurrentAccount(102, 5000, "Current", 3000);

System.out.println("==== SAVINGS ACCOUNT ====");

    savings.displayAccount();

System.out.println("\nDepositing Rs.2000");

    savings.deposit(2000);

System.out.println("\nWithdrawing Rs.1500");

    savings.withdraw(1500);

System.out.println("\nCalculating Interest");

    savings.calculateInterest();

System.out.println("\nUpdated Savings Account:");

    savings.displayAccount();

System.out.println("\n==== CURRENT ACCOUNT ====");

    current.displayAccount();

System.out.println("\nDepositing Rs.3000");

    current.deposit(3000);

System.out.println("\nWithdrawing Rs.9000");

    current.withdraw(9000);

System.out.println("\nUpdated Current Account:");

    current.displayAccount();

System.out.println("\n==== TRANSFER ====");
```

```
System.out.println("Transferring Rs.1000 from Savings to Current");

    savings.transfer(current, 1000);

System.out.println("\nSavings Account after transfer:");

    savings.displayAccount();

System.out.println("\nCurrent Account after transfer:");

    current.displayAccount();

sc.close();

}

}
```

Compilation & Execution Commands :

Save the file exactly as : BankAccountManagement.java

Compile the program: javac BankAccountManagement.java

Run the program: java BankAccountManagement

Output Screenshot :

```
PS C:\JavaPrograms> java BankAccountManagement
===== SAVINGS ACCOUNT =====
Account Number : 101
Account Type   : Savings
Balance       : 10000.0

Depositing Rs.2000
Amount deposited successfully.

Withdrawing Rs.1500
Amount withdrawn successfully.

Calculating Interest
Interest Rate : 5.0%
Interest Added: 525.0

Updated Savings Account:
Account Number : 101
Account Type   : Savings
Balance       : 11025.0

===== CURRENT ACCOUNT =====
Account Number : 102
Account Type   : Current
Balance       : 5000.0
```

Result :

Thus, the Bank Account Management System was successfully implemented using classes, objects, constructors, inheritance, method overriding, interest calculation, overdraft facility, and account transfer in Java.